

# German erased at a tenth of the collateral — English protected, Romance undefended

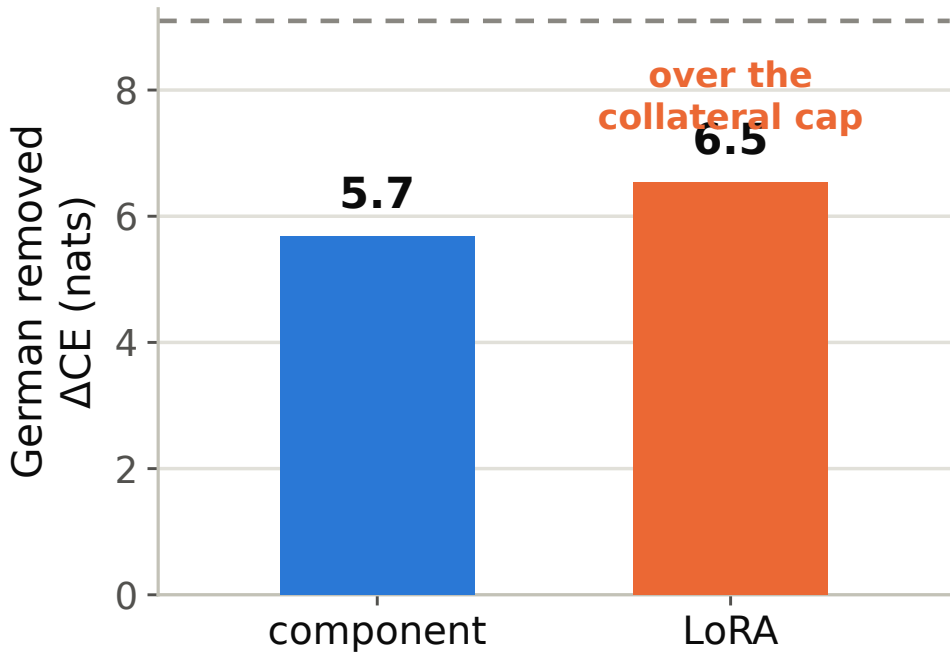
Nothing defends French, Spanish or Italian. Component: the most German it can remove while holding Romance under 0.10 nats.

LoRA at its strongest: over the full sweep of (learning rate, lambda, training step), whichever setting removes at least as much

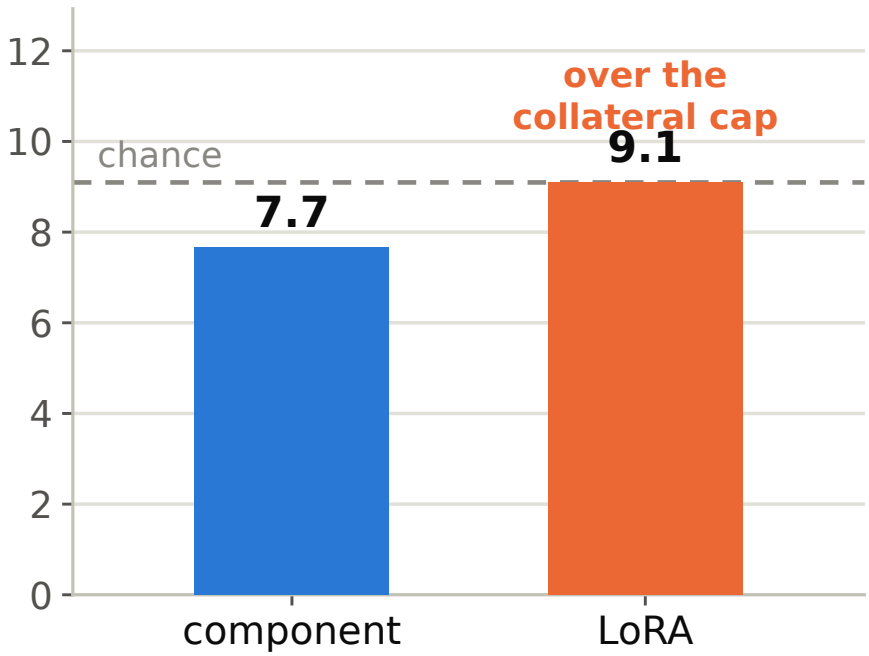
German as the component and does the least damage doing it. Chosen on dev/guard blocks, reported on held-out blocks.

■ single component c3634 (112 scalars) ■ LoRA r=1 (704,512 parameters) - - German at chance (+9.10)

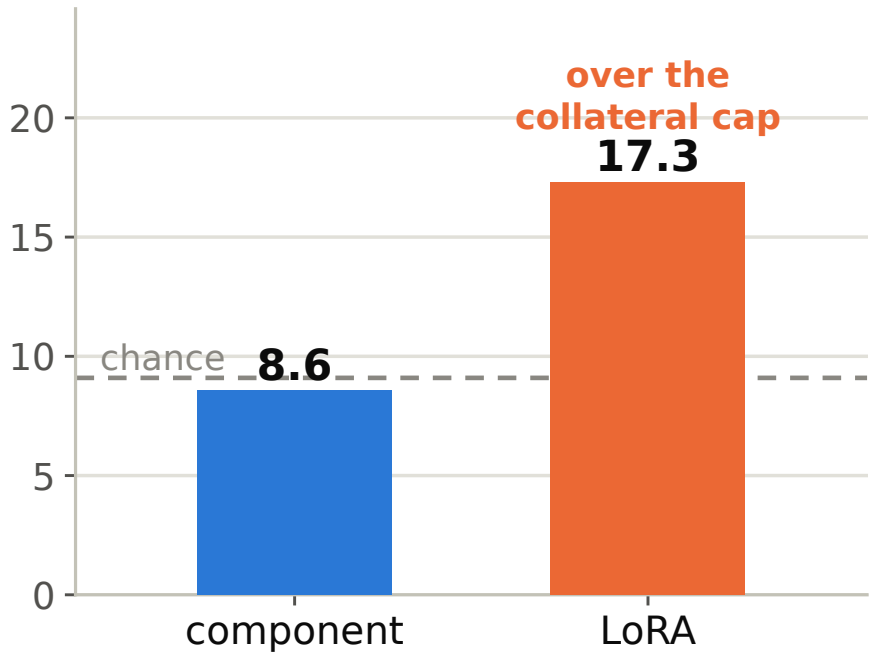
8 German tokens



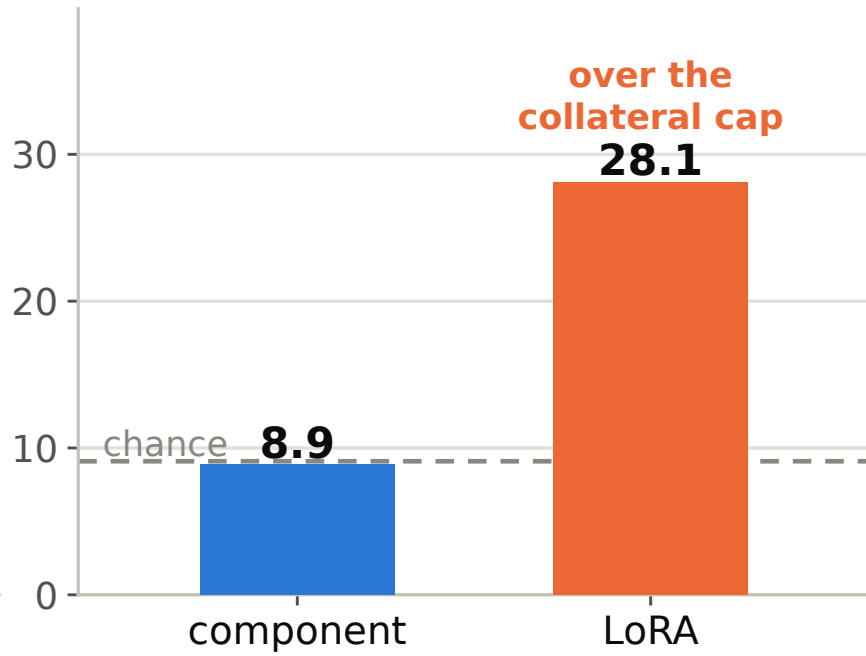
64 German tokens



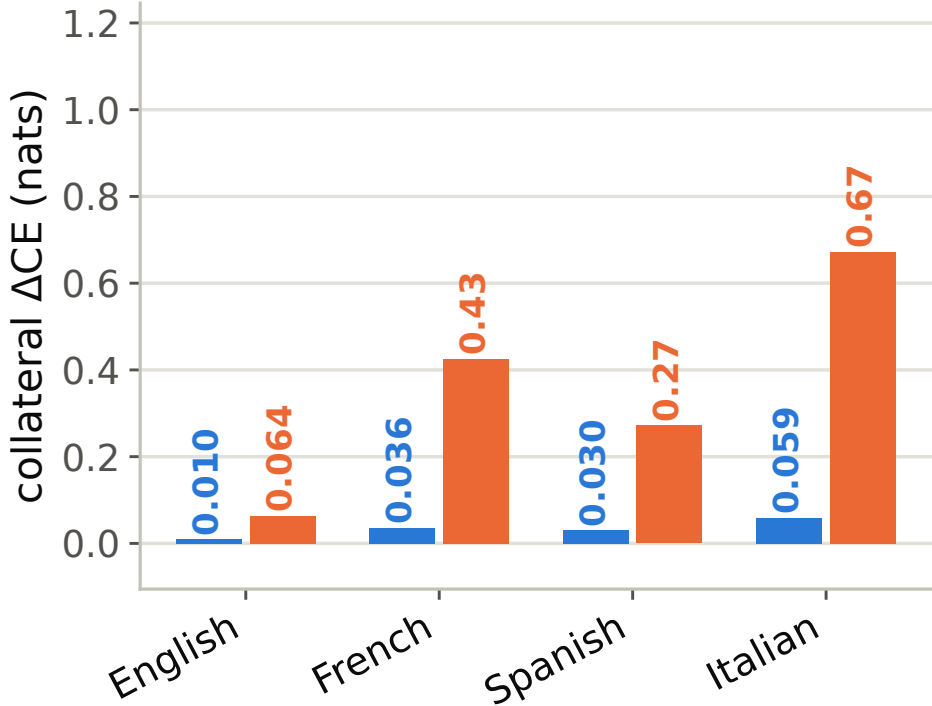
512 German tokens



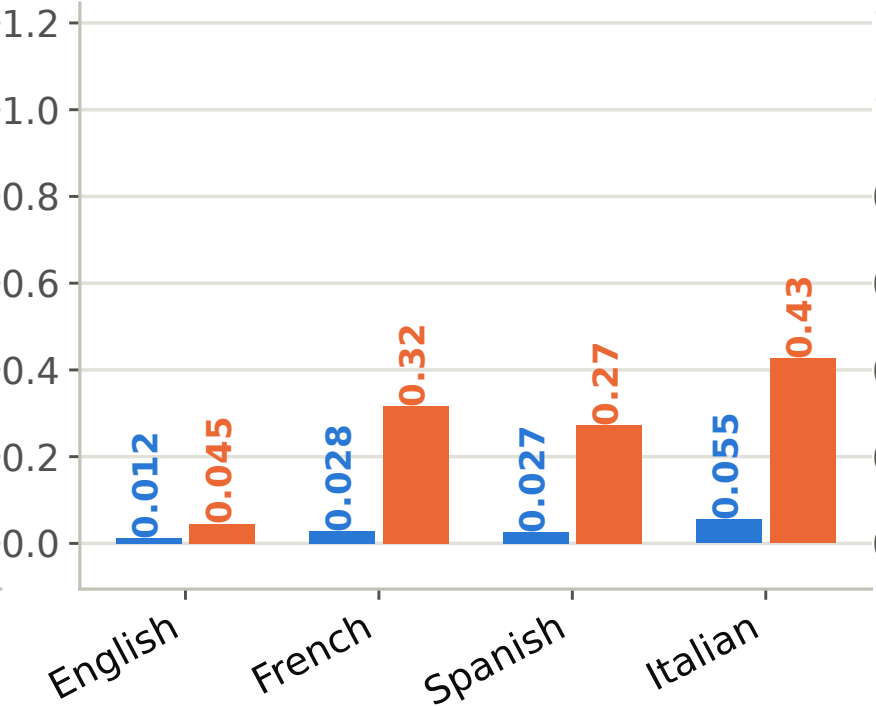
2,048 German tokens



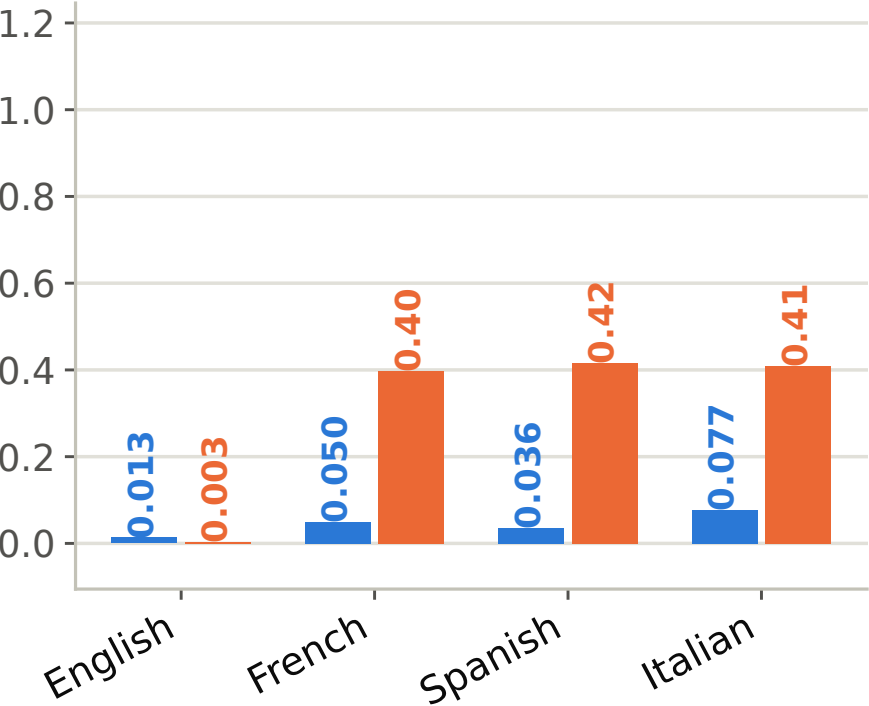
worst: 0.06 vs 0.67



worst: 0.06 vs 0.43



worst: 0.08 vs 0.42



worst: 0.08 vs 0.88

